

Data Source
Jurkat Perturb-seq
(ARC Virtual Cell Challenge)

jurkat.h5
21,412 cells x 18,080 genes

Control + 68
Perturbations

1. Preprocessing

extract highly variable genes (3051)

Log-normalization (provided)

Encode perturbation labels (68 + control)

Construct input features
 $x = [\text{control_expression} \mid \text{one_hot_perturb}]$

Output: Processed Matrix X, Labels y

2. Feature Engineering

Control Cells as input (3051)

One-hot perturbation vector (68)

Concatenate: [3051 genes + 68 pert] = 3119

Target: Perturbed Cells (3051)

Input dim: 3119 | Output dim: 3051

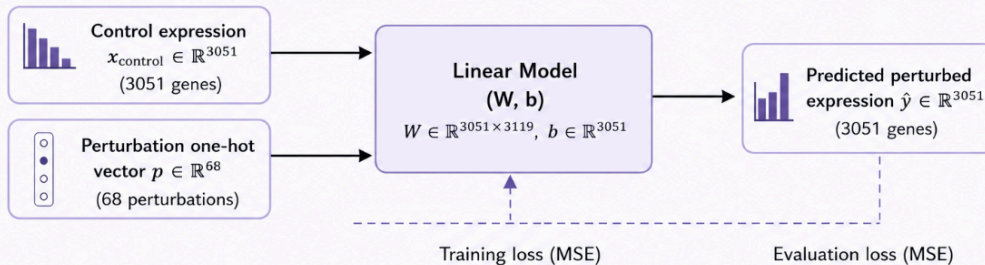
3. Data Splitting (by Perturbation Group)

Train Set
Remaining
cells (~80%)

Validation Set
From pert. with
>60 cells
(10% of each
group)

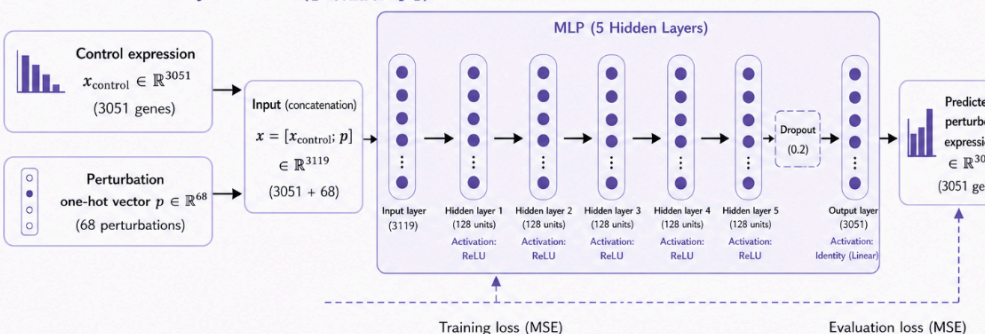
Test Set
From pert.
groups
(10% each
group)

4.1 Linear Baseline: $\hat{y} = Wx + b$



Loss function: **MSELoss**. Optimizer: **Adam** Learning rate: **1e-4, 1e-3, 1e-2**
Batch size: **32, 64, 128**, Weight decay: **0.0, 1e-5, 1e-4**

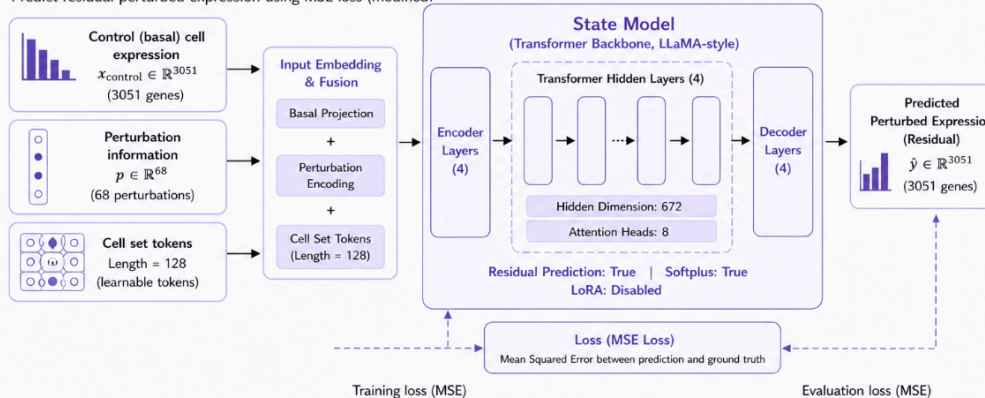
4.2 MLP Model: $\hat{y} = \text{MLP}([x_{\text{control}}; p])$



Loss function: **MSELoss**. Optimizer: **Adam** Learning rate: **1e-4, 1e-3, 1e-2**
Batch size: **32, 64, 128**, Dropout rate: **0.2 and 0**

4.3 State Model (Transformer-based) (modified)

Predict residual perturbed expression using MSE loss (modified)



Loss function: **MSELoss**. Optimizer: **Adam** Learning rate: **1e-3** Batch size: **64**

5. Result

State Model has the lowest training loss and validation loss.

Linear Model Summary

Total runs	27
Learning rate	1e-4
Batch size	32
Weight decay	1e-4
Training loss	0.371
Validation loss	0.373599
Test loss	0.365355

MLP Model Summary

Total runs	18
Learning rate	1e-3
Hidden dimension	128
Dropout	0.2
Batch size	64
Training loss	0.353 (at epoch 10)
Validation loss	0.356352 (best)
Test loss	0.355626

State Model Summary

Learning rate	1e-3
Hidden dimension	128
Batch size	64
Training loss	0.327
Validation loss	0.334

6. Tools



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